Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

PENDAHULUAN
TINJAUAN PUSTAKA
macros-ohp fontawesome lipsum multimedia setspace graphicx

Lindsey the Tour Guide Robot

Name Surname

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Lincoln Centre for Autonomous Systems, UoL, UK

L-CAS away day, somewhere in Lincoln,
September 2019

Presentation Content

Project Overview

Usage Patterns in Long-Term Deployment

Current Work & Outlook

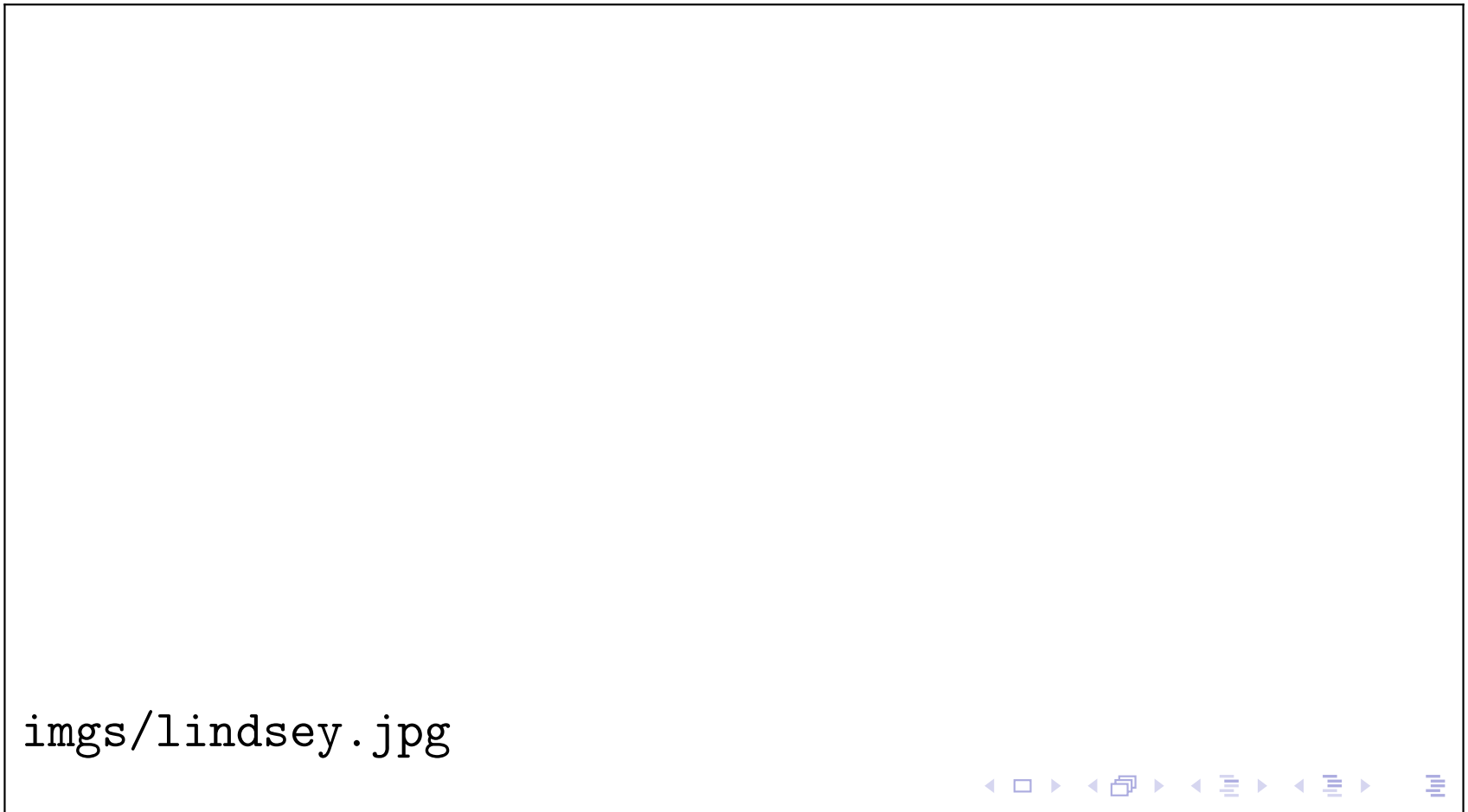
The project

7cm(5.5cm,0.5cm)

Lindsey the robot

Autonomous robot that gives guided tours and information to the public at The Collection Museum in Lincoln.

7.5cm(5cm,3cm)



The project

Objectives

1. Learn to improve/adapt the robot social capabilities (its behaviors) from real world experience while interacting with museum visitors.
2. Increase the public engagement level when interacting with the robot.

From: *Lindsey the Tour Guide Robot-Usage Patterns in a Museum Long-Term Deployment. F Del Duchetto, P Baxter, M Hanheide. International Conference on Robot & Human Interactive Communication (RO-MAN) 2019*

Robot Operations

Guided tours

Lindsey guides the visitors through a sequence of 5/6 exhibits linked by a common theme, describing each of them verbally and with images.

Go to exhibit and describe

Lindsey guides the visitors to an exhibit chosen from the map and describes its content with two different difficulty level.

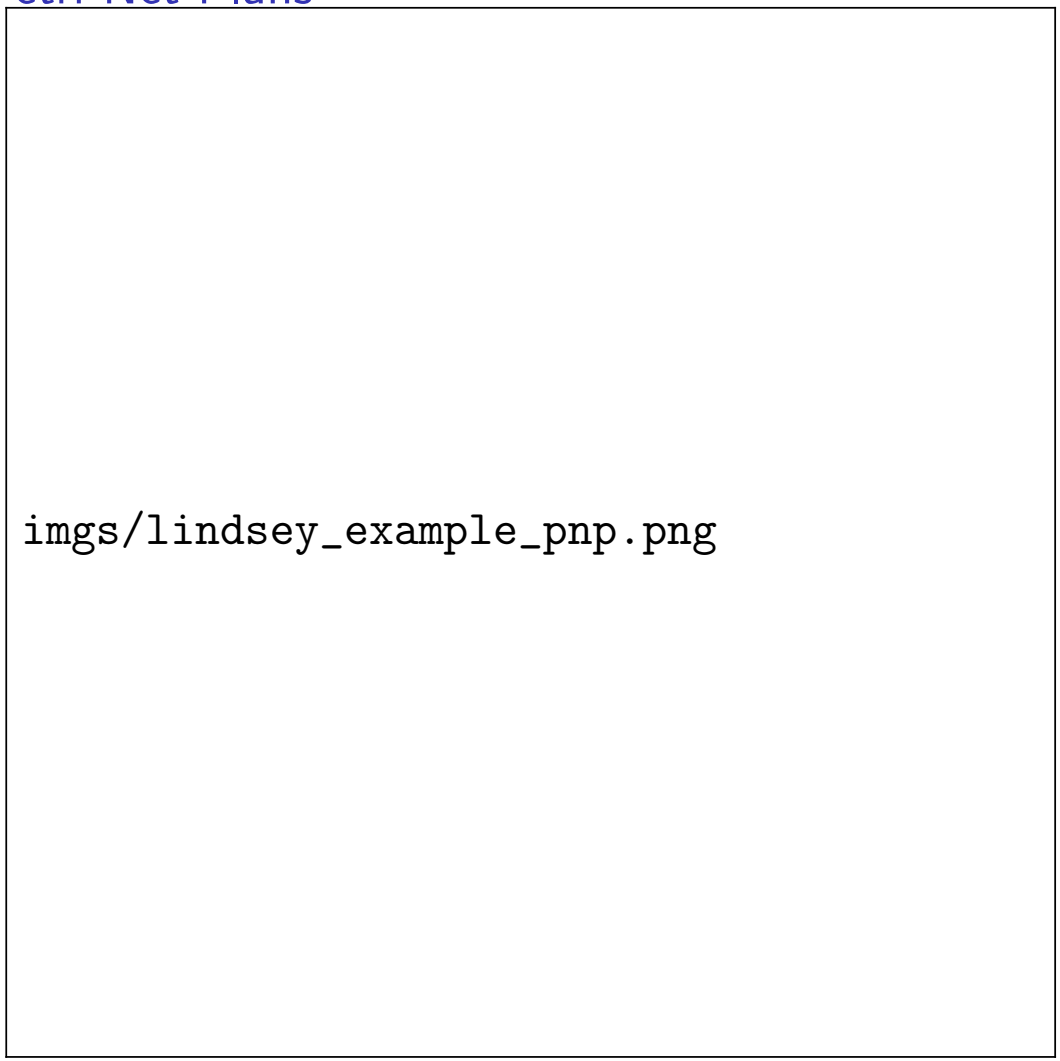
Describe exhibit

The robot gives a short verbal description of the exhibit requested by the visitor.

Robot Framework

Behaviors specification

Petri Net Plans

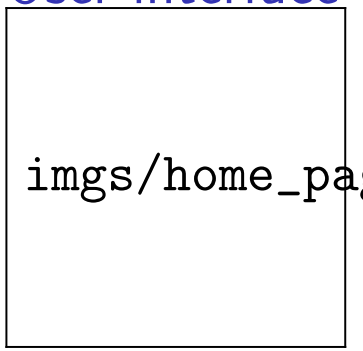


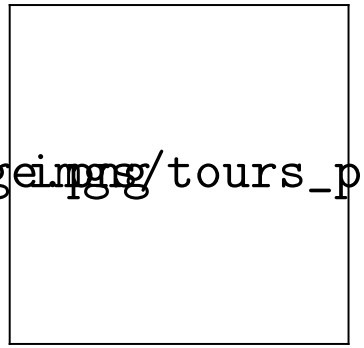
- Conditional plans made of **actions**, **conditions** and **Execution Rules**.
- A PNP can be translated into a stochastic policy.

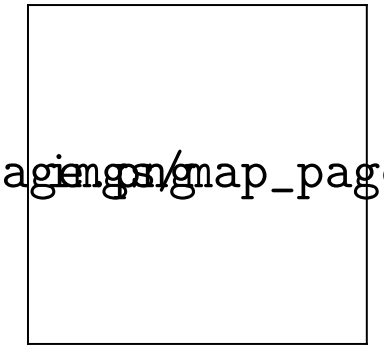
Robot Framework

Robot Management

User interface

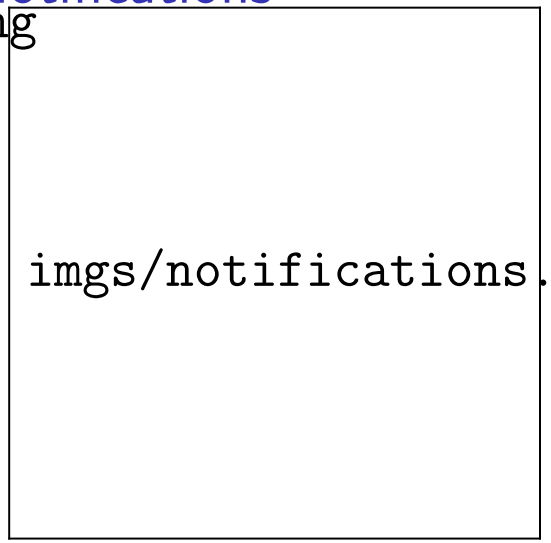




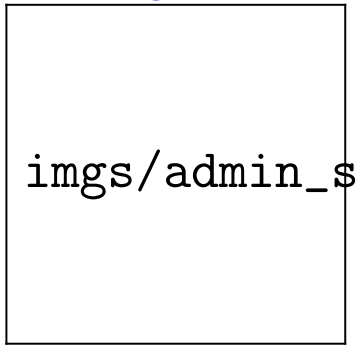


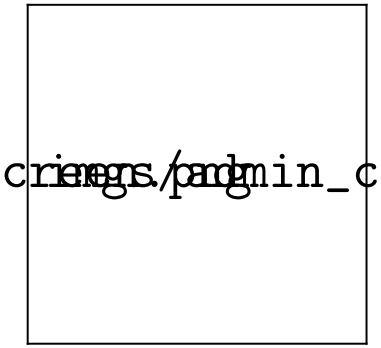
https://github.com/laurencejbelliott/roswebcomponents

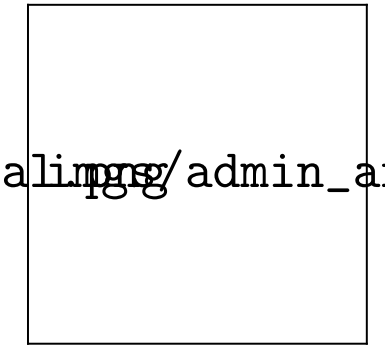
Critical events notifications



Management interface







Data Analysis

Tabel: Long-Term Autonomy metrics.

Days of operation	103 days
Total distance travelled	299 km
Total tasks completed	8423
TLS	26 days, 11 hours
A%	74%

Tabel: Number of user demanded tasks with their duration.

Task	Tot. domanded	Average duration	Median duration	Shortest	Longest
Guided tour	2367	4.52 [min]	3.13 [min]	11 [sec]	22.15 [min]
Go to exhibit and describe	2379	1.87 [min]	1.84 [min]	8 [sec]	30.79 [min]
Describe exhibit	489	26.73 [sec]	20.03 [sec]	7.31 [sec]	5.78 [min]

4.4cm(5mm, 26.5mm) How the tasks ends?

imgs/tasks_fate.png

5cm(75mm, 30mm) What's the tours duration?

imgs/tours_duration.png

Discussion

Engaged by the robot?

Engagement is easily started with Lindsey, but it is typically lost after 2 minutes.

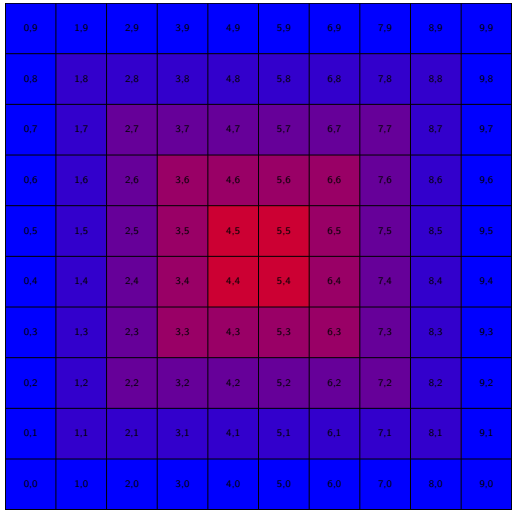
How does a tour guide engage the public?

From: Katie Best. Making museum tours better: Understanding what a guided tour really is and what a tour guide really does.

- tours should not resemble monolithic lectures but they must be **interactive**;
- guides should facilitate **audience contribution** through questions and answers;
- guides should seek to secure audience attention to inform and entertain them, encouraging them to **orient to the feature under consideration**;
- the audience should not be considerate as a whole but the guide must take into account features of the single people, **personalizing the experience**;
- technologists need to create non-human guides that have a similar level of **sensitivity to the audience** built-in.

Learning from engagement level

[66]



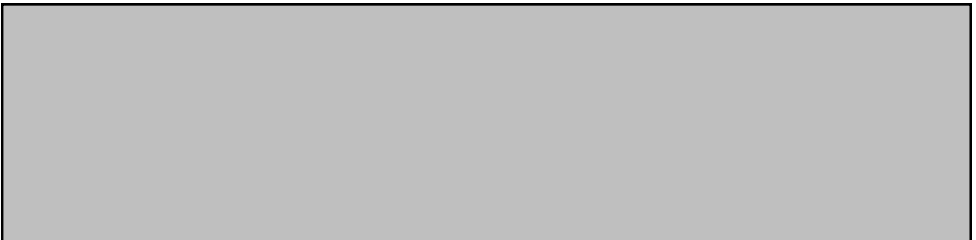
Engagement model

Annotations

[width=height=0.8poster, autostart, borderwidth=2pt, loop, open, repeat, showcontrols=true]Video not found, or use Okularvids/15588759624761199 5_annotations.mp4

- 3 Annotators
- Continuous value for each frame in [0,1]
- Inter-rater Spearman correlation ρ: 0.82

Engagement model



Model predictions

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Engagement model

Annotations

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Model predictions

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Thank you for your attention!

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